Forces and friction

Mathematics · Mechanics Year 2

Chapter 5

Forces and friction

Resolving forces · Inclined planes · Limiting friction

Overview

Resolving forces

Replace a force by perpendicular components. Use these to find a resultant or apply Newton's second law.

Inclined planes

Choose axes parallel and perpendicular to the plane.

Friction

Distinguish friction below its limiting value from limiting friction.

Unless stated otherwise, use $g = 9.8\,\mathrm{m\,s^{-2}}$.

A displacement of 5 m makes 60° above the horizontal. Find its horizontal and vertical components.

Displacement

A force of 7 N makes 60° to the upward vertical, towards the right. Resolve it.

Force

Tip: the component adjacent to the stated angle uses cosine; the opposite component uses sine.

Quickfire Questions 4

Resolve 8 N, 30° above the positive horizontal.

a

Resolve 6 N, 45° above the negative horizontal.

b

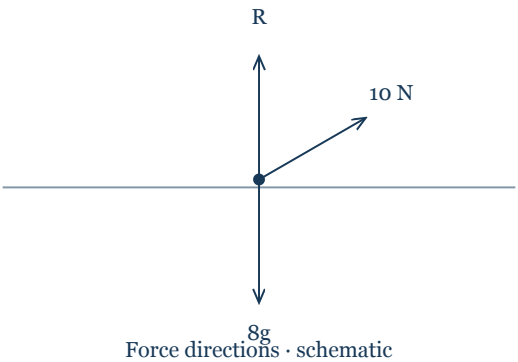
Resolve 9 N, 37° below the positive horizontal.

c

Applied Example 5

A box of mass 8 kg rests on a smooth horizontal floor. A force of 10 N pulls at 30° above the horizontal. Find the acceleration and normal reaction.

Answer

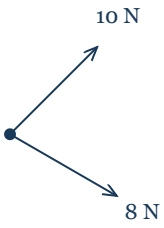


Force directions · schematic

Combining Forces 6

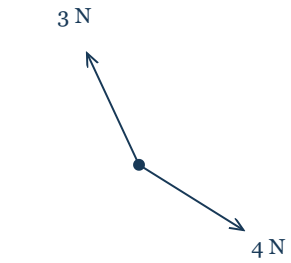
Find the resultant of 10 N at 45° above the positive horizontal and 8 N at 30° below it.

Answer



Force directions · schematic

Find the resultant of 3 N, 25° left of the upward vertical, and 4 N, 32° below the positive horizontal.



Force directions · schematic

Answer

Exercise 5A · Question 4

Three forces are in equilibrium. Angles below are measured anticlockwise from the positive horizontal. Find the magnitude and direction of B.

(a) 15 N at 150°; 20 N at 210°.

a

(b) 25 N at 50°; 10 N at −30°.

b

(c) 10 N at 120°; 20 N at −20°.

c

Exercise 5A · Question 6

A 10 kg particle on a smooth horizontal surface is pulled by P at 45° above the horizontal. Its acceleration is $2\,\text{m}\,\text{s}^{-2}$. Find P. (3 marks)

6

Question 7

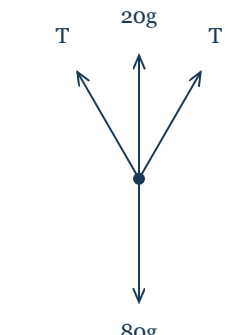
A particle of mass m on a smooth horizontal surface is pulled by 20 N at 25° above the horizontal. Its acceleration is $0.5\,\text{m}\,\text{s}^{-2}$. Find m. (3 marks)

7

Exercise 5A · Question 8

A parachutist of mass 80 kg descends at constant velocity. Two supporting lines each make 60° with the horizontal and have tension T. Air resistance on the parachutist is one quarter of the weight. Find T.

Answer

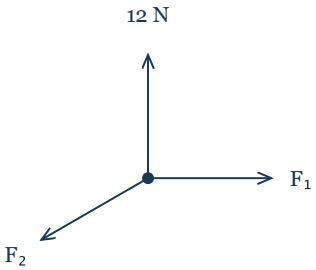


Force directions · schematic

Exercise 5A · Question 9

Forces act at a point: 12 N vertically upwards, F_1 horizontally right, and F_2 at 30° below the negative horizontal. Their resultant is $(2\sqrt{3}\mathbf{i} + 2\mathbf{j})$ N. Find F_1, F_2.

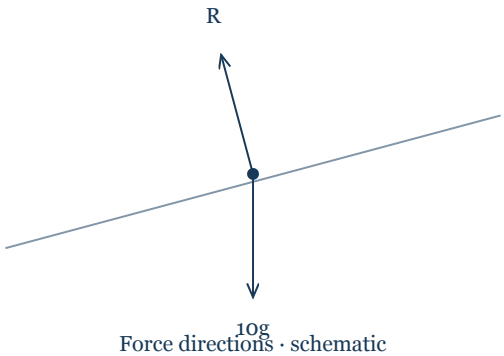
Answer



Force directions · schematic

A particle of mass 10 kg slides down a smooth plane inclined at 15°. Find the normal reaction and acceleration.

Answer

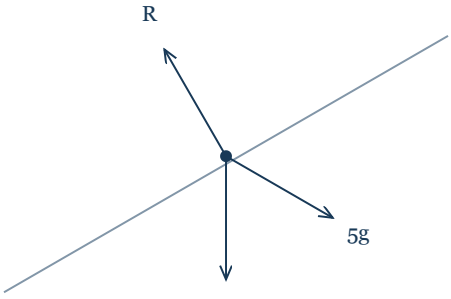


Force directions · schematic

Inclined Plane with an additional force 13

A particle of mass m is pushed up a smooth 30° slope by a force of $5g$ N. The force makes 60° with the upward slope, directed into the plane. The acceleration is $0.5\,\mathrm{m\,s^{-2}}$ up the slope. Find m.

Answer

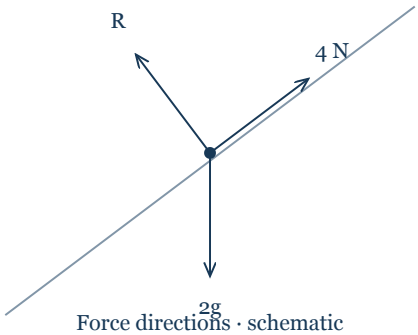


Force directions · schematic

Test Your Understanding 14

A 2 kg particle on a smooth plane is pulled by 4 N up the plane. The inclination α satisfies $\tan\alpha = 3/4$. Find the acceleration.

Answer



Force directions · schematic

Exercise 5B · Question 4

A 6 kg particle is pulled up a rough 15° slope by a 30 N force parallel to the plane, at constant velocity.

(a) Draw a force diagram. (4 marks)

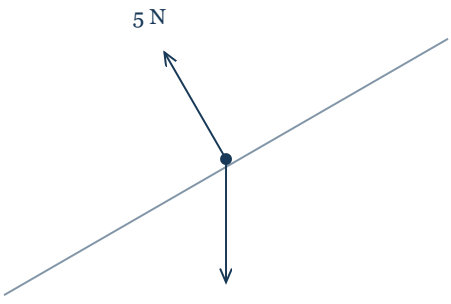
(b) Find the frictional force. (5 marks)

Force diagram

Answer

Exercise 5B · Question 5

A particle of mass m slides down a smooth 30° plane. The normal reaction is 5 N. Find (a) m (3 marks); (b) its acceleration (3 marks).



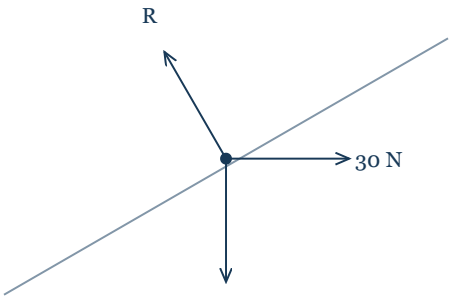
Force directions · schematic

Answer

Exercise 5B · Question 6

A 5 kg particle on a smooth 30° slope is pushed by a horizontal 30 N force towards the rising slope. Find its acceleration.

Answer

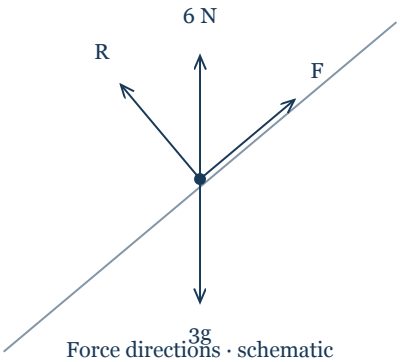


Force directions · schematic

Exercise 5B · Question 7

A 3 kg particle on a rough 40° slope has a vertically upward force of 6 N applied. It moves at constant velocity. Find the frictional force.

Answer

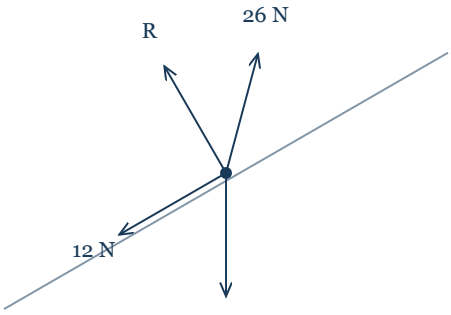


Force directions · schematic

Exercise 5B · Question 8

A particle of mass m moves up a rough slope with $\tan\alpha = 1/\sqrt{3}$. A 26 N force acts 45° above the plane. Friction is 12 N down the plane and the acceleration is $1\,\mathrm{m\,s^{-2}}$ up the plane. Find m to 3 significant figures. (3 marks)

Answer



Force directions · schematic

Friction adjusts to the situation

1. No horizontal force on a stationary object on a horizontal surface: $F = 0$.

2. Pull T on a stationary object: friction balances T, until the limiting value is reached.

3. At limiting equilibrium, or sliding in the model used here:

$F = \mu R$.

Friction opposes motion, or the tendency to move.

Below the limit:

$0 \leq F \leq \mu R$.

Do not automatically use $F = \mu R$ for every stationary particle.

Coefficient of friction

$F_{\max} = \mu R$.

μ is dimensionless and describes the pair of surfaces.

R is the normal reaction, not automatically the weight.

Rougher surfaces generally have a larger coefficient. A larger normal reaction increases the limiting friction.

On an inclined plane, resolve perpendicular to the plane first. Applied forces can change R.

μ need not be less than 1.

Illustrative static coefficients from the source

Steel on steel: 0.74; aluminium on steel: 0.61.

Copper on steel: 0.53; rubber on concrete: 1.0.

Wood on wood: 0.25–0.5; glass on glass: 0.94.

Waxed wood on wet snow: 0.14.

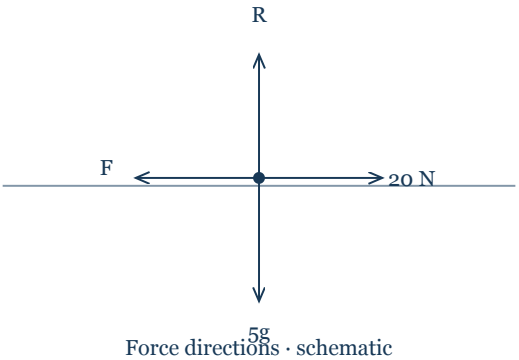
Ice on ice: 0.10; Teflon on Teflon: 0.04.

Synovial joints: 0.01.

No value was supplied for waxed wood on dry snow or lubricated metal on metal.

A 5 kg particle is pulled along a rough horizontal surface by a horizontal force of 20 N. The coefficient of friction is 0.2. Find its acceleration.

Answer

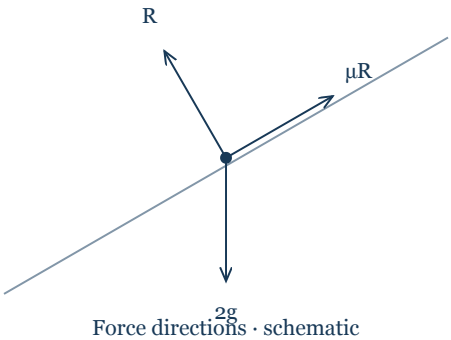


Force directions · schematic

Inclined Plane with Friction Example 23

A 2 kg particle slides down a rough 30° plane with acceleration $1\,\mathrm{m\,s^{-2}}$. Find μ.

Answer



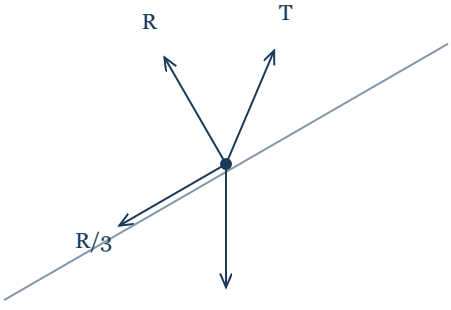
Force directions · schematic

Test Your Understanding 24

Edexcel M1 June 2013 · Question 3

A box of mass 2 kg rests on a rough plane inclined at 30°. A light inextensible rope pulls at angle α above the plane, where $\tan\alpha = 3/4$. The coefficient of friction is $1/3$. The box is on the point of moving up the plane. Model it as a particle and find the tension. (8 marks)

Answer



Force directions · schematic

Exercise 5C · Question 3

A 0.5 kg particle slides down a rough 15° slope with acceleration $0.25\,\text{m}\,\text{s}^{-2}$. Find μ to 3 significant figures. (3 marks)

3

Question 4

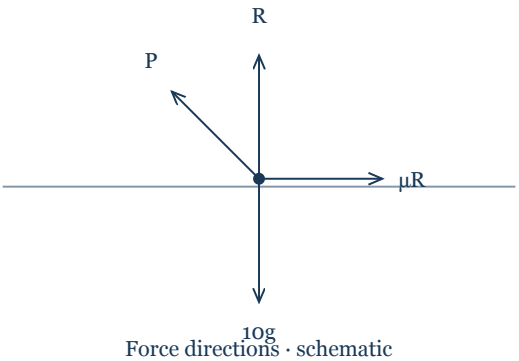
A 2 kg particle slides down a rough 20° plane. A force P acts up the plane. Given $\mu = 0.3$ and acceleration $0.2\,\text{m}\,\text{s}^{-2}$ down the plane, find P. (4 marks)

4

Exercise 5C · Question 6

A sled of mass 10 kg is pulled across rough horizontal ground by a force P at 45° above the horizontal towards the left. Given $\mu = 0.1$ and acceleration $0.3\,\text{m}\,\text{s}^{-2}$ left, find P.

Answer



Force directions · schematic

Exercise 5C · Question 7

A train travelling at $20\,\text{m}\,\text{s}^{-1}$ stops in $30\,\text{s}$ after its wheels lock.

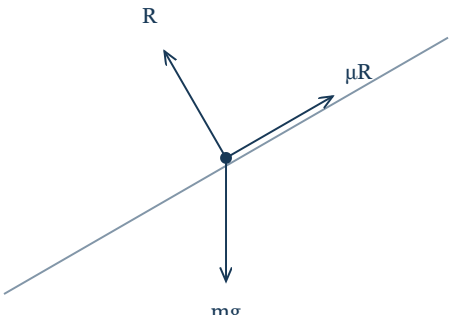
(a) Model the train as a particle and assume friction is the only resistance. Find μ.

a

(b) How would allowing for air resistance change the inferred coefficient? (2 marks)

b

A particle is on the point of sliding down a rough plane inclined at θ. Prove that $\mu = \tan\theta$. Can μ exceed 1?



Force directions · schematic

Answer